Simón Rodríguez Arango

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Education

- **B.S. in Astronomy (5-year professional degree)**, Universidad de Antioquia — 2021 – Present (Expected: Dec. 2026)
  - **Undergraduate Thesis:** Beyond LTE: Evaluation of HCN and C_2H_2 as truncation tracers in highly irradiated protoplanetary disks.
  - **Cumulative GPA:** 4.23/5.0 (passing grade: 3.0).
  - **Selected advanced coursework:** Astrochemistry (5.0), Radio Astronomy (5.0), Modern Astrophysics (4.9), Stellar Astrophysics, Galactic and Extragalactic Astrophysics, Astrostatistics.
- **B.S. in Systems Engineering (5 semesters completed)**, Universidad de Antioquia — 2018 – 2021

Research Interests

Protoplanetary disk chemistry and planet formation; exoplanet atmospheres; JWST/MIRI spectroscopy.

Publications

- Frediani, J., Getman, K. V., Woitke, P., et al. (incl. **Rodríguez, S.**; 14th of 21 authors) (2026). "XUE. ProDiMo models of internally and externally irradiated planet-forming disks around 0.3–4.0 $M_\odot$ stars (The IRIS project I)." Submitted to A&A. arXiv:2609.08621.

Research Experience

- **Undergraduate Researcher (B.S. Thesis)** | Universidad de Antioquia | Advisor: Prof. Germán Chaparro — 10/2025 – Present
  - Member of the XUE collaboration, a JWST/MIRI survey of protoplanetary disks in massive star-forming regions.
  - Developed custom Python codes to estimate non-LTE level populations of mid-infrared molecular transitions, and analyzed thermochemical models from ProDiMo (Woitke, Kamp & Thi 2009) and synthetic spectra computed with FLiTs (Woitke et al. 2018) at different truncation radii.
- **Research Assistant – Subsurface water stability on Mars** | Universidad de Antioquia | Advisor: Prof. Jorge Zuluaga — 05/2025 – 10/2025
  - Developed a numerical model in Python to solve hydrostatic equilibrium equations and generate global pressure and temperature profiles in the Martian crust, and analyzed data from the InSight mission.

Relevant Academic Projects

- **Project: Observational Astronomy and Data Reduction - Practical Astronomy II Course** — 2024
  - Executed a complete pipeline for the reduction and calibration of science images using IRAF.
  - Performed aperture photometry in I and V bands for the subsequent analysis of color-magnitude diagrams.
- **Project: Exoplanet Detection and Characterization - Planetary Sciences Course** — 2024

  - Extracted and processed radial velocity and photometry data from confirmed exoplanet catalogs.
  - Developed Python routines to construct Doppler and transit curves, identifying instrumental parameters.
- **Project: Mock Observing Proposal for a Dark Galaxy Candidate - Radio Astronomy Course** 2026
  - Wrote a full-format JVLA HI 21 cm proposal (science case, target selection from the FASHI catalog, configuration and sensitivity justification) to map the gas distribution and kinematics of a starless HI source.

Conference Presentations

- **Rodríguez, S.**, & Chaparro, G. (2026). "Beyond LTE: Evaluation of HCN and C_2H_2 as truncation tracers in highly irradiated protoplanetary disks." Oral talk presented at the IX Colombian Congress of Astronomy and Astrophysics (COCOA 2026), Bogotá, Colombia.
- Numpaque, S., Montoya, B., **Rodríguez, S.**, & Pérez, D. (2025). "Mapping thermodynamic conditions for subsurface water stability on Mars." Poster presented at the National Congress of Physics (Colombian Physical Society), Manizales, Colombia.

Technical Skills

- **Programming Languages & Databases:** Python (NumPy, SciPy, Matplotlib, Pandas, Astropy, ProDiMoPy), SQL, Java.
- **Software & Tools:** ProDiMo, FLiTs, IRAF, DS9, Git/GitHub, LaTeX, Linux/Unix environments.
- **Languages:** English (Fluent), Spanish (Native).

Complementary Education

- **LATAM Hack Days "Open Tools, Open Sky" (Rubin/LSST)** | Universidad de Antioquia 10/2026
  - Worked in a team developing a pipeline to filter Vera C. Rubin Observatory alerts from community brokers and cross-match supernova candidates with their host galaxies in archival catalogs.
- **1st Intl. Program in Observational Astronomy (virtual)** | O.R.B.I.T.A. 07/2026
  - Observed with the Light Bridges robotic telescopes at Teide Observatory (Spain), working with real data under the mentorship of international researchers.

Leadership and Service

- **Member of the RECA Mentorship Committee** 09/2025 – Present
  - Manage communication logistics and track student attendance for panels and program activities.
  - Analyze data and generate statistics from participation forms to evaluate the program's impact.